Emily Qi

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Education

University of Waterloo 2024 – 2028
Candidate for Bachelor of Computer Science, Honours (3.9 GPA)
- Computing and Financial Management Outstanding Achievement Award

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, HTML, CSS, SQL
Libraries and Frameworks: React, Next.js, Node.js, Express, Django, FastAPI, Flask, PostgreSQL, MongoDB, SQLite, OpenAI, OpenCV, PyTorch, scikit-learn, pandas, NumPy, Selenium
Tools: Git, Docker, Firebase, Google Cloud, Cloudflare, Vercel, Railway, Weights & Biases, Postman, Figma

Experience

Machine Learning Engineering Intern Jan 2026 – present
HubHead
- Developed a visual grounding validation system for 250+ document types: generates field descriptions from bounding box annotations via Gemini 2.5 Pro, predicts entity positions from text alone, and iteratively refines via IoU against ground-truth for per-edge spatial feedback
- Built a multimodal template recommender using Gemini 768-dim embeddings with statistical ranking (z-score normalization, clarity-weighted aggregation, consensus scoring), adaptive subsampling, and <1.0s cached inference
- Scaled ML extraction pipelines to 70K+ pages across Document AI, Gemini, and Textract on Cloud Run, offloading to a dedicated worker service with streaming ingestion, semaphore-gated concurrency, heartbeat fault recovery, and real-time team collaboration via Flask-SocketIO

Machine Learning Researcher Sep 2025 – Dec 2025
University of Waterloo
- Built an end-to-end Soft Actor-Critic (SAC) reinforcement learning pipeline for autonomous racing in Trackmania with Stable Baselines 3, OpenAI Gym, and TMRL's OpenPlanet telemetry interface
- Designed a multi-component reward function combining track progress, speed incentives, and action smoothness penalties with tunable weights, and tracked training runs with Weights & Biases

Software Engineering Intern May 2025 – Aug 2025
Royal Canadian Mounted Police
- Drove end-to-end development of a full-stack app (React, Next.js, FastAPI, SQLite) using RESTful APIs and CRUD operations for asynchronous execution of key operational workflows, reducing manual overhead by 70%
- Led client conversations to translate business needs into product requirements, accelerating delivery by 2 months
- Engineered robust browser automation using Selenium to navigate a dynamic, bot-resistant web interface

Projects

Karyon | *Django, React, OpenAI, sentence-transformers, OpenCV, Vercel, Railway, Docker, PostgreSQL*
- Developed full-stack **video Q&A platform** enabling natural language queries with timestamp-linked citations and multimodal processing (audio, visual, hybrid), supporting uploads and YouTube imports
- Engineered a RAG pipeline combining Whisper transcription, GPT-4o vision keyframe analysis, and semantic chunking with sentence-transformer embeddings; used OpenCV change detection to reduce vision API calls by 80%
- Containerized and deployed to production with JWT auth, Fernet-encrypted API keys, and per-user data isolation

Atari Breakout with DQN | *Python, PyTorch, Deep Reinforcement Learning, OpenCV, Weights & Biases*
- Implemented and combined three deep RL papers (DQN, Double DQN, Prioritized Experience Replay) from scratch in PyTorch; trained agent for 10M frames on Atari Breakout, achieving 7× reward improvement and 227× Q-value growth over random baseline
- Built memory-efficient prioritized replay buffer with uint8 frame compression (4× storage reduction), TD-error prioritization, and importance sampling bias correction for stable training on constrained GPU hardware

Biquadris | *C++, Object-Oriented Programming, X11*
- Architected a C++ **multiplayer Tetris game** applying 5 GoF design patterns across a 15+ class polymorphic hierarchy, enabling composable block modifiers, decoupled dual-display rendering (text + X11) synchronized real-time, and a macro-supported command system with recursion safeguards